ArithmeticSite

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Chapter 1

The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of $\mathbb{Q}$. Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

Definition 1.1. Let $\mathbb{N}^\times$ denote the set $\{1, 2, 3, \ldots\}$ of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that $\mathbb{N}^\times$ is the free commutative monoid on the set of primes: every element factors uniquely as a finite product $\prod_p p^{a_p}$. This factorisation structure is not merely background — it will resurface as the combinatorial backbone of the site structure on $B\mathbb{N}^\times$, where factorizations of integers into coprime parts give rise to covering families whose overlap conditions reflect divisibility.

Definition 1.2. The *classifying category* $B\mathbb{N}^\times$ of $\mathbb{N}^\times$ is the one-object category whose single object is $*$ and whose morphisms $* \to *$ are the elements of $\mathbb{N}^\times$, with composition given by multiplication.

Remark 1.3. The passage from a monoid M to the one-object category BM is the standard *delooping* construction. A presheaf on BM — that is, a functor $BM^{\mathrm{op}} \to \mathbf{Set}$ — is exactly a set X equipped with a right action of M: each element $n \in M$ acts as a function $X \to X$, and the action is compatible with composition. The presheaf topos $\widehat{\mathbb{N}^\times}$ introduced in Chapter 3 is precisely the category of such sets-with-$\mathbb{N}^\times$-action, and it is in this topos that the Arithmetic Site lives.

Chapter 2

The tropical semiring $\bar{\mathbb{N}}$

Definition 2.1. The *tropical semiring* $\bar{\mathbb{N}}$ is the set $\mathbb{N} \cup \{+\infty\}$ equipped with:

- *Tropical addition*: $a \oplus b := \min(a, b)$, with identity element $+\infty$.
- *Tropical multiplication*: $a \odot b := a + b$ (ordinary addition), with identity element 0.

In Lean, $\bar{\mathbb{N}}$ is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e. $+\infty$) and the multiplicative identity is `trop 0`.

Lemma 2.2. *$\bar{\mathbb{N}}$ is a commutative semiring.*

Proof. This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

Lemma 2.3. *Tropical addition on $\bar{\mathbb{N}}$ is idempotent: $a \oplus a = a$ for all $a \in \bar{\mathbb{N}}$.*

Proof. $a \oplus a = \min(a, a) = a$. In Lean this follows from `Tropical.add_eq_left`. □

Definition 2.4. For each $n \in \mathbb{N}^\times$, the *Frobenius endomorphism* $\varphi_n : \bar{\mathbb{N}} \to \bar{\mathbb{N}}$ is defined by

$$\varphi_n(x) := n + x \quad (x \in \mathbb{N}), \qquad \varphi_n(+\infty) := +\infty,$$

where $+$ denotes ordinary addition in $\mathbb{N} \cup \{+\infty\}$. Equivalently, in `Tropical (WithTop ℕ)`, this is tropical multiplication by `trop (n : WithTop ℕ)`, i.e. $\varphi_n(x) = \mathtt{trop}\, n \odot x$.

In Lean, φ_n is defined as an `AddMonoid.End NBar` via `frobeniusEndomorphism`.

Lemma 2.5. *For each $n \in \mathbb{N}^\times$, the map $\varphi_n : \bar{\mathbb{N}} \to \bar{\mathbb{N}}$ is an additive monoid endomorphism, i.e. it preserves tropical addition and the zero element $+\infty$:*

$$\begin{aligned} \varphi_n(+\infty) &= +\infty, \\ \varphi_n(a \oplus b) &= \varphi_n(a) \oplus \varphi_n(b). \end{aligned}$$

The second identity follows from the distributivity $n + \min(a, b) = \min(n + a, n + b)$.

Remark 2.6. Note that φ_n is *not* a semiring homomorphism in general, since it does not preserve the multiplicative identity: $\varphi_n(0) = n \neq 0$ for $n > 0$. It is however a homomorphism of $\bar{\mathbb{N}}$-modules.

Lemma 2.7. *The Frobenius endomorphisms satisfy the composition law*

$$\varphi_m \circ \varphi_n = \varphi_{m+n}$$

for all $m, n \in \mathbb{N}^\times$. *In particular, the map* $n \mapsto \varphi_n$ *is a semigroup homomorphism from* $(\mathbb{N}^\times, +)$ *into the endomorphism semigroup* $\mathrm{End}_{\mathrm{AddMon}}(\bar{\mathbb{N}})$.

Proof. For any $x \in \bar{\mathbb{N}}$: $(\varphi_m \circ \varphi_n)(x) = m + (n + x) = (m + n) + x = \varphi_{m+n}(x)$. In Lean this is `frobenius_comp`, proved via `mul_assoc` in `Tropical (WithTop ℕ)`. □

Chapter 3

The presheaf topos $\widehat{\mathbb{N}^\times}$

Definition 3.1. The *presheaf topos* $\widehat{\mathbb{N}^\times}$ is the category of functors $B\mathbb{N}^{\times\,\mathrm{op}} \to \mathbf{Set}$, i.e. the category of sets equipped with a left $\mathbb{N}^\times$-action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

Lemma 3.2. $\widehat{\mathbb{N}^\times}$ *is a Grothendieck topos.*

Proof. Any presheaf category on a small category is a Grothendieck topos. □

Chapter 4

The structure sheaf

Definition 4.1. The *structure sheaf* $\mathcal{O}$ of the Arithmetic Site is the object of $\widehat{\mathbb{N}^\times}$ whose underlying set is $\bar{\mathbb{N}}$ and whose $\mathbb{N}^\times$-action is given by the Frobenius endomorphisms: $n \cdot x := \varphi_n(x)$.

Lemma 4.2. *The assignment* $n \mapsto \varphi_n$ *defines a semigroup homomorphism* $(\mathbb{N}^\times, +) \to \mathrm{End}_{\mathrm{AddMon}}(\bar{\mathbb{N}})$, *making* $\bar{\mathbb{N}}$ *into a left* $(\mathbb{N}^\times, +)$*-module.*

Chapter 5

The Arithmetic Site

Definition 5.1. The *Arithmetic Site* is the pair $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ consisting of the presheaf topos $\widehat{\mathbb{N}^\times}$ together with its structure sheaf $\mathcal{O}$.

Chapter 6

Points of the topos

Definition 6.1. A *point* of a topos $\mathcal{E}$ is a geometric morphism $p : \mathbf{Set} \to \mathcal{E}$, i.e. an adjoint pair $p^* \dashv p_*$ where p^* is left exact.

Theorem 6.2. *The points of* $\widehat{\mathbb{N}^\times}$ *are in natural bijection with the adèle class space* $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \hat{\mathbb{Z}}^*$.

Proof. **(sorry)** $\square$

Chapter 7

The Frobenius correspondences

Definition 7.1. For each $\lambda \in \mathbb{Q}_+^\times$, the *Frobenius correspondence* Φ_λ is a correspondence on $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$ arising from the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space.

Theorem 7.2. *The Frobenius correspondences realize the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space at the level of the topos $\widehat{\mathbb{N}^\times}$.*

Proof. **(sorry)** $\square$

Chapter 8

Connection to the Riemann zeta function

Theorem 8.1. *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function $\zeta(s)$.*

Proof. **(sorry)** $\square$